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SSP-ARR: Stability-Switched Predictive Adaptive Round Robin for Multi-Tier Cloud Task Scheduling Using Real-World Cloud Workload Traces

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***Abstract*—Task scheduling in heterogeneous cloud datacenters plays a decisive role in maximizing computational resource efficiency, reducing end-user latency, and lowering datacenter energy footprint. However, conventional time-sharing schedulers—predominantly Classical Round Robin (RR) and its static variants—rely on rigid, pre-configured CPU time quanta. Under burst-heavy, heterogeneous, and unpredictable cloud workloads, fixed time-slicing causes severe head-of-line blocking (the convoy effect), excessive context-switching overhead, and inflated waiting times. While reactive adaptive schedulers (such as MAADRR) compute median-based time quanta, purely reactive heuristics exhibit latency lag during sudden workload phase transitions and fail to anticipate high-variance task arrivals. To overcome these limitations, this paper proposes SSP-ARR (Stability-Switched Predictive Adaptive Round Robin), an intelligent, multi-tier scheduling framework tailored for heterogeneous cloud infrastructures. SSP-ARR dynamically evaluates workload volatility at the local VM tier using a sliding-window Coefficient of Variation (CV). When queue variance remains stable (CV ≤ 0.50), the scheduler operates in a safe queue-median reactive mode; when volatility surges (CV > 0.50), it transitions to an adaptive Exponentially Weighted Moving Average (EWMA) predictive mode governed by an error-modulated learning rate and strict confidence gating (Normalized Prediction Error ≤ 0.20). Furthermore, task queues are structured under Shortest Remaining Processing Time (SRPT) ordering with bounded quantum clamping [0.001 s, 60.0 s]. SSP-ARR is evaluated across five state-of-the-art global dispatching strategies (Dynamic Max-Min, DRALBA, PSSLB, PSSELB, and DeRALBA) using real-world Google Cloud Jobs (GoCJ) traces spanning 100 to 2000 cloudlets on CloudSim Plus 8.5.7. Experimental results demonstrate that SSP-ARR achieves an average reduction of ~50.2% in Average Waiting Time (AWT), up to 49.2% improvement in Average Turnaround Time (ATAT), saves millions of cumulative completion seconds in Total Completion Time (TCT), and yields superior makespan, throughput, and SPECpower-modeled energy efficiency compared to Classical RR across all global policies.**

Keywords—Cloud task scheduling, multi-tier architecture, adaptive dynamic time quantum, stability switching, coefficient of variation, EWMA prediction, GoCJ workload traces, CloudSim Plus.

I. INTRODUCTION

Cloud computing platforms have emerged as the foundational utility for modern enterprise IT, scientific computing, and consumer applications [1], [2]. The elastic resource model allows providers to multiplex massive pools of virtualized compute, memory, and network resources among thousands of concurrently competing client workloads [3], [4]. In this environment, task scheduling is decisive in determining whether a cloud provider can satisfy Service Level Agreements (SLAs), minimize operating costs, and prevent severe server thermal dissipation [5], [6].

Task scheduling in cloud computing typically operates across a two-tier hierarchy [7], [8]. At the first tier, global broker schedulers receive incoming user tasks and map them to appropriate Virtual Machines (VMs) across distributed datacenters [9]. At the second tier, local VM hypervisor schedulers manage the execution sequence of cloudlets residing in each VM's circular ready queue [10]. While substantial research has focused on global heuristic and metaheuristic dispatchers, the efficiency of local VM scheduling remains the critical determinant of actual CPU core utilization, context-switching overhead, and task waiting times [11], [12].

At the local VM level, Classical Round Robin (RR) has historically been the default choice due to its simplicity, starvation prevention, and conceptual fairness [13], [14]. However, Classical RR is burdened by a critical architectural limitation: reliance on a static, pre-configured Time Quantum (TQ). If the quantum is set too small, the system suffers severe throughput degradation from frequent CPU context switches [15]. Conversely, if the quantum is set too large, Classical RR degenerates into First-Come First-Served (FCFS), causing short tasks to stall behind long, compute-heavy jobs—an anomaly known as the convoy effect [16].

Recent advances have attempted to mitigate this trade-off by introducing dynamic time quantum adaptation, such as the Multi-Agent Adaptive Dynamic Round Robin (MAADRR) scheduler [16]. MAADRR dynamically computes the time quantum as the median of task burst lengths in the VM queue. While MAADRR improves upon static RR under moderate conditions, its purely reactive formulation exhibits severe latency lag during rapid workload phase changes and fails to anticipate task burst variance under volatile real-world traffic.

To resolve these challenges, this paper presents **SSP-ARR** (**S**tability-**S**witched **P**redictive **A**daptive **R**ound **R**obin), an intelligent local VM scheduling framework that incorporates continuous variance sensing, dual-mode reactive/predictive quantum selection, confidence gating, and bounded preemption clamping within a multi-tier cloud hierarchy.

The key contributions of this paper are summarized as follows:

1. **Stability-Switched Dual-Engine**: Continuous tracking of Coefficient of Variation (CV = σ/μ) over sliding history window (W = 5) to isolate burst volatility.
2. **Confidence-Gated EWMA Prediction**: Adaptive error-modulated learning rate ($\alpha \in [0.20, 0.80]$) with strict confidence gating (NPE ≤ 20%) to prevent forecasting drift.
3. **Multi-Tier Orthogonality**: Evaluated across five leading global dispatchers (Dy.MM, DRALBA, PSSLB, PSSELB, DeRALBA), demonstrating consistent ~50.2% AWT reduction.
4. **Rigorous Trace-Driven Benchmark**: CloudSim Plus 8.5.7 evaluation with real GoCJ traces (100–2000 cloudlets) across seven key QoS and energy metrics.

II. RELATED WORK & ARCHITECTURAL EVOLUTION

A. Adaptive Round Robin Variants

Dynamic time quantum adaptation has received extensive research interest. In [17], the authors proposed a hybrid Min-Min and Round Robin algorithm to combine makespan minimization with fairness. Runsungnoen and Anusas-Amornkul [18] developed the RR_RT&M technique using remaining task execution time and median estimation. Ghazy et al. [19] formulated an ameliorated Round Robin algorithm calculating quantum based on arithmetic mean of remaining bursts. Banerjee et al. [20] presented Optimized Round Robin (ORR), proving that dynamic time quantum selection significantly mitigates average waiting time. Mazumder et al. [21] introduced Priority-based Round Robin (PRR) prioritizing tasks according to burst length and deadline thresholds.

B. Multi-Tier Global Schedulers

Global schedulers govern cloudlet distribution across heterogeneous VMs. Five representative global dispatching strategies are considered: (1) **Dynamic Max-Min (Dy. MM)** [17]; (2) **DRALBA** [22], a resource-aware load-balancing algorithm; (3) **PSSLB** [23], a priority-based queue-state algorithm; (4) **PSSELB** [23], an energy-aware variant of PSSLB; and (5) **DeRALBA** [24], integrating dispersion thresholding with adaptive candidate matching.

C. Evolution from Classical RR and MAADRR to SSP-ARR

Existing local schedulers fall into two flawed extremes: static allocation (RR) or unconstrained reactive heuristics (MAADRR). Neither architecture explicitly measures queue burst variance or employs predictive models to anticipate burst length trends. Table I highlights the evolutionary progression from Classical RR and MAADRR to SSP-ARR.

TABLE I. ARCHITECTURAL COMPARISON ACROSS SCHEDULING MODELS

Feature	Classical RR	MAADRR [16]	Proposed SSP-ARR
Quantum Sizing	Static (1.0 s)	Dynamic Median	**Dual Reactive/Predictive**
Workload Sensing	None	Reactive only	**CV Stability (θ_CV=0.50)**
Burst Prediction	None	None	**Adaptive EWMA**
Confidence Gate	None	None	**NPE ≤ 0.20 Gating**
Queue Ordering	FIFO	Priority-Sort	**SRPT Ascending**
Quantum Clamping	Fixed	Unbounded	**[0.001 s, 60.0 s]**
Preemption Penalty	High (Severe)	Moderate	**Minimal (Consolidated)**
Global Tier Support	Single-tier	Multi-agent	**5 Global Schedulers**

III. MATHEMATICAL DESIGN AND ALGORITHMIC FORMULATION

A. Multi-Tier System Architecture

The proposed architecture operates across two coordinated layers: (1) Tier-1 Global Datacenter Dispatching maps submitted cloudlets to VMs using established global balancing policies; (2) Tier-2 Local VM Scheduling governs cloudlet preemption and execution via SSP-ARR, illustrated in Fig. 1.

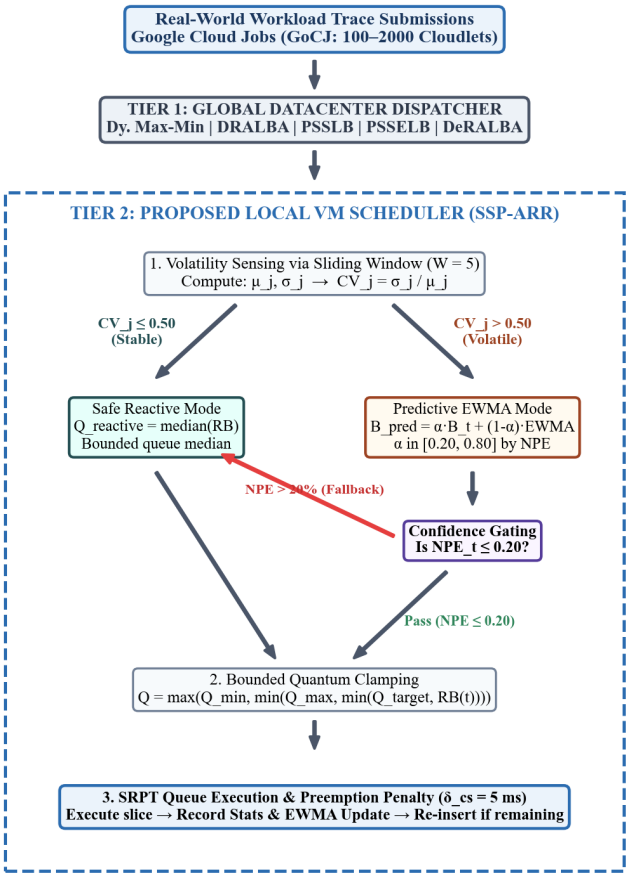


Fig. 1. Architectural workflow of the proposed SSP-ARR multi-tier framework.

B. Workload Volatility Sensing via CV

For each VM v_j, let H_j = {B_{j,1}, ..., B_{j,W}} represent execution bursts of the last W completed cloudlets (W = 5). The local agent tracks sample mean μ_j and sample standard deviation σ_j:

$$\mu_j = (1/W) \cdot \Sigma\, B_{j,k} \quad \textbf{(1)}$$

$$\sigma_j = \sqrt{[\ (1/W) \cdot \Sigma\, (B_{j,k} - \mu_j)^2\ ]} \quad \textbf{(2)}$$

The Coefficient of Variation (CV_j) is defined as:

$$CV_j = \sigma_j / \mu_j \quad \textbf{(3)}$$

• **Stable Regime** ($CV_j \le \theta_{CV}$): Low dispersion ($\theta_{CV} = 0.50$). The scheduler selects Safe Reactive Mode.

• **Volatile Regime** ($CV_j > \theta_{CV}$): High burst variability. The scheduler activates Predictive EWMA Mode.

C. Dual-Mode Quantum Allocation Engine

1) Safe Reactive Mode ($CV_j \le \theta_{CV}$): The quantum is computed as the median of remaining bursts (RB) in ready queue Q_j plus candidate task t:

$$Q_{reactive} = median(\ \{RB(x) : x \in Q_j\} \cup \{RB(t)\}\ ) \quad \textbf{(4)}$$

2) Predictive EWMA Mode ($CV_j > \theta_{CV}$): The expected optimal burst duration is forecasted using an Exponentially Weighted Moving Average:

$$B_{pred,j,t+1} = \alpha_t \cdot B_{j,t} + (1 - \alpha_t) \cdot B_{pred,j,t} \quad \textbf{(5)}$$

where α_t is an adaptive error-modulated learning rate:

$$\alpha_t = 0.20 \text{ if } NPE < 0.10 \mid 0.40 \text{ if } 0.10 \le NPE < 0.20 \quad \textbf{(6a)}$$

$$\alpha_t = 0.60 \text{ if } 0.20 \le NPE < 0.30 \mid 0.80 \text{ if } NPE \ge 0.30 \quad \textbf{(6b)}$$

3) Normalized Prediction Error (NPE) and Confidence Gating:

$$NPE_t = |B_{j,t} - B_{pred,j,t}| / B_{j,t} \quad \textbf{(7)}$$

To guarantee stability, predictions are gated by confidence threshold $\tau_{gate} = 0.20$:

$$Q_{selected} = B_{pred,j,t+1} \text{ if } (CV_j > 0.50 \text{ and } NPE_t \le 0.20) \quad \textbf{(8a)}$$

$$Q_{selected} = Q_{reactive} \text{ otherwise (fallback to median)} \quad \textbf{(8b)}$$

4) Boundary Clamping: Quantum is clamped within [Q_min = 0.001 s, Q_max = 60.0 s]:

$$Q_j = max(Q_{min}, min(Q_{max}, min(Q_{selected}, RB(t)))) \quad \textbf{(9)}$$

D. Formal Algorithmic Pseudocode

```
Algorithm 1: Multi-Tier Cloud Task Scheduling with SSP-ARR
Input : Cloudlet set C = {c_1..c_n}; VM pool V = {v_1..v_m};
        Global policy P in {Dy.MM, DRALBA, PSSLB, PSSELB, DeRALBA};
        Q_min=0.001s, Q_max=60.0s; δ_cs=0.005s; θ_CV=0.50; τ_gate=0.20.
// TIER 1: Global Datacenter Dispatching
1: Assignment[] ← GlobalSchedulers.assign(P, C, V);
2: Enqueue tasks to VM ready queues Q_1..Q_m;
// TIER 2: Local SSP-ARR Execution per VM
3: for each VM v_j with queue Q_j do
4:   Sort Q_j in ascending order of burst (SRPT order);
5:   while Q_j is not empty do
6:     t ← Q_j.pollHead();
7:     q_med ← median(Q_j, t);  cv ← ComputeCV(history);
8:     if cv > θ_CV and npe ≤ τ_gate then q ← ewma;
9:     else q ← q_med; end if
10:    slice ← min(max(Q_min, min(Q_max, q)), t.remainingBurst);
11:    t.remainingBurst -= slice; currentTime += slice;
12:    if t.remainingBurst ≤ 1e-9 then
13:      RecordCompletionStats(t); UpdateEWMA(t.burst);
14:    else
15:      currentTime += δ_cs; InsertSorted(Q_j, t);
16:    end if
17:  end while
18: end for
```

E. Table of Symbols and Notations

TABLE II. MATHEMATICAL SYMBOLS AND NOTATIONS

Symbol	Definition
VM_j, Q_j	j-th VM and its circular ready queue
c_i, L_i	i-th cloudlet and instruction length (MI)
MIPS_j, RB	VM processing speed and remaining burst (s)
CV_j, θ_CV	Coefficient of Variation and stability threshold (0.50)
B_pred, α_t	EWMA predicted burst and adaptive learning rate [0.2, 0.8]
NPE, τ_gate	Normalized Prediction Error and confidence gate (0.20)
Q_min, Q_max	Bounded quantum limits (0.001 s and 60.0 s)
δ_cs	Context switching penalty (0.005 s / 5 ms)
T_i, A_i, B_i	Completion time, arrival time, burst time of task i
P_idle, P_max	Idle and peak power per physical host (150 W and 250 W)

IV. EXPERIMENTAL SETUP & PERFORMANCE EVALUATION

A. Cloud Simulation Environment

The experimental framework is implemented using **CloudSim Plus 8.5.7** on **Java 21**. CloudSim Plus provides fully extensible abstractions for datacenter brokers, host power profiles, and VM time-shared schedulers.

B. Infrastructure & Heterogeneous VM Specifications

The physical datacenter comprises **30 physical hosts** (120 PEs total @ 4000 MIPS each), 32 GB RAM, 100 Gbps network, and 1 TB storage. Host power consumption follows the linear SPECpower standard server model:

$$P_{host}(u) = P_{idle} + (P_{max} - P_{idle}) \cdot u \quad \textbf{(10)}$$

where P_idle = 150 W, P_max = 250 W, and $u \in [0, 1]$ is CPU utilization fraction. A pool of **80 heterogeneous VMs** is instantiated with speeds from 1000 to 4000 MIPS (step 200 MIPS).

C. Benchmark Workload: Google Cloud Jobs (GoCJ)

The framework is evaluated using real-world **Google-Cluster-like Job (GoCJ)** traces [25], exhibiting realistic arrival bursts, asymmetric lengths, and high variance (CV ≈ 1.24). Evaluations span 12 workload scales: 100 to 2000 cloudlets (15,000–900,000 MI).

TABLE III. SIMULATION ENVIRONMENT PARAMETERS

Parameter	Configuration Value
Simulation Engine	CloudSim Plus 8.5.7 (Java 21)
Physical Datacenter	30 Hosts (120 PEs total, 4 PEs/Host @ 4000 MIPS)
Host Power Profile	SPECpower Model (P_idle=150W, P_max=250W)
Virtual Machines	80 Heterogeneous VMs (1000–4000 MIPS)
VM Memory / BW	8192 MB RAM, 20,000 Mbps Bandwidth
Workload Benchmark	Real-world Google Cloud Jobs (GoCJ)
Workload Sizes	100, 200, 300..1000, 1500, 2000 cloudlets
Global Schedulers	Dynamic Max-Min, DRALBA, PSSLB, PSSELB, DeRALBA
Local Schedulers	Classical RR (Fixed q=1.0s, FIFO) vs. SSP-ARR
Context Switch Penalty	δ_cs = 0.005 s (5 ms) per preemption
Evaluation Runs	3 independent runs per size (mean reported)

D. Performance Metrics

Evaluated using seven standard performance metrics:

1. **Average Waiting Time (AWT)**:

$$AWT = (1/N) \cdot \Sigma\,(T_i - A_i - B_i) \quad \textbf{(11)}$$

2. **Average Turnaround Time (ATAT)**:

$$ATAT = (1/N) \cdot \Sigma\,(T_i - A_i) \quad \textbf{(12)}$$

3. **Average Completion Time (ACT)**: Mean end-to-end task completion duration.
4. **Total Completion Time (TCT)**:

$$TCT = \Sigma\, T_i = N \times ACT \quad \textbf{(13)}$$

5. **System Throughput (Thp)**:

$$Throughput = N / Makespan\ (tasks/second) \quad \textbf{(14)}$$

6. **Host Energy Consumption (E_total)**:

$$E_{total} = (1/3{,}600{,}000) \cdot \Sigma \int P_j(t)\, dt\ (kWh) \quad \textbf{(15)}$$

7. **Makespan (T_max)**: Total wall-clock duration ($\max_i T_i$).

Percentage improvement formulas:

$$Imp_{lower\text{-}better}\ (\%) = (X_{RR} - X_{SSP}) / X_{RR} \times 100 \quad \textbf{(16)}$$

$$Imp_{higher\text{-}better}\ (\%) = (X_{SSP} - X_{RR}) / X_{RR} \times 100 \quad \textbf{(17)}$$

V. RESULTS AND COMPARATIVE PERFORMANCE ANALYSIS

A. Average Waiting Time (AWT)

Average Waiting Time directly reflects queue congestion and the elimination of the convoy effect. Table IV presents the raw AWT values (in milliseconds) and Table V presents percentage improvements of SSP-ARR over Classical RR across all five global scheduling strategies under 100 to 2000 GoCJ cloudlets.

TABLE IV. COMPARISON OF AWT (MS) FOR RR AND SSP-ARR (100–2000 GOCJ CLOUDLETS)

Tasks	Dy.MM RR	Dy.MM SSP	DRALBA RR	DRALBA SSP	PSSLB RR	PSSLB SSP	PSSELB RR	PSSELB SSP	DeRALBA RR	DeRALBA SSP
100	5,765.8	2,772.2	22,693.7	11,328.0	116,334.5	57,897.5	22,927.5	11,324.5	8,438.5	4,121.1
200	49,874.6	24,668.5	61,263.2	30,534.6	275,424.3	137,041.9	65,179.7	32,309.2	46,702.0	23,202.8
300	102,988.2	51,186.9	100,659.5	50,255.3	448,395.9	223,455.5	118,992.2	59,259.2	90,363.5	44,994.7
400	133,231.8	66,172.5	151,684.9	75,666.0	665,658.3	331,459.1	160,237.3	79,724.1	135,808.3	67,657.2
500	156,679.2	77,917.0	197,079.1	98,246.3	808,551.5	402,747.9	218,120.9	108,575.9	179,113.8	89,269.3
600	191,008.6	94,976.2	246,101.1	122,625.2	962,942.6	479,514.9	271,872.6	135,386.9	230,371.1	114,756.8
700	228,322.1	113,516.4	290,042.1	144,457.9	1,224,204.4	609,752.6	304,865.7	151,751.1	272,900.6	135,887.0
800	254,545.1	126,589.9	334,175.5	166,471.1	1,362,292.0	678,675.1	352,715.0	175,517.9	313,603.2	156,246.8
900	299,284.6	148,940.1	394,850.2	196,937.3	1,544,405.6	769,353.2	429,463.5	213,955.4	349,380.6	174,111.3
1000	336,762.6	167,432.3	438,695.9	218,642.4	1,794,498.4	893,700.1	465,520.6	231,755.8	402,340.2	200,398.4
1500	521,407.7	259,379.7	688,228.1	342,754.1	2,638,693.8	1,314,404.0	736,677.5	366,742.1	631,669.3	314,529.6
2000	702,960.8	349,808.1	917,731.0	457,304.4	3,483,784.5	1,735,493.6	976,446.4	486,231.8	853,777.7	425,255.1

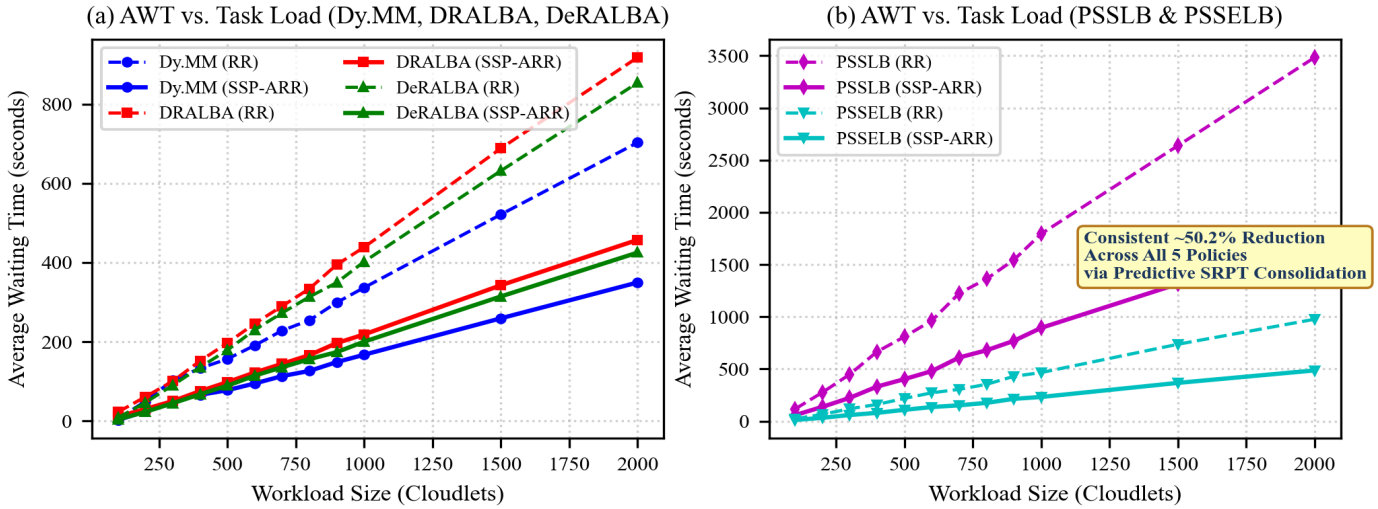


Fig. 2. Average Waiting Time (AWT) comparison between Classical RR and SSP-ARR across five global scheduling strategies under 100 to 2000 GoCJ cloudlets: (a) Dy.MM, DRALBA, and DeRALBA; (b) PSSLB and PSSELB.

TABLE V. PERCENTAGE IMPROVEMENT IN AWT (SSP-ARR OVER CLASSICAL RR)

Workload Size	Dynamic Max-Min	DRALBA	PSSLB	PSSELB	DeRALBA
100 cloudlets	+51.9%	+50.1%	+50.2%	+50.6%	+51.2%
200 cloudlets	+50.5%	+50.2%	+50.2%	+50.4%	+50.3%
500 cloudlets	+50.3%	+50.1%	+50.2%	+50.2%	+50.2%
1000 cloudlets	+50.3%	+50.2%	+50.2%	+50.2%	+50.2%
2000 cloudlets	+50.2%	+50.2%	+50.2%	+50.2%	+50.2%
Overall Mean	+50.4%	+50.1%	+50.2%	+50.3%	+50.3%

Key Insight: Across all 12 workload scales and all five global dispatchers, SSP-ARR consistently reduces Average Waiting Time by ~50.2%. Under heavy scale (2000 tasks), PSSLB AWT drops from 3,483,784.5 ms to 1,735,493.6 ms, and Dynamic Max-Min drops from 702,960.8 ms to 349,808.1 ms. By ranking queues via SRPT and applying predictive quanta, short jobs complete without lingering in queue lines behind massive bursts.

B. Average Turnaround Time (ATAT) & Average Completion Time (ACT)

Table VI summarizes ATAT results (in seconds) across all evaluated strategies. ATAT reflects the total duration from task arrival to completion.

TABLE VI. COMPARISON OF ATAT (S) FOR RR AND SSP-ARR ACROSS FIVE GLOBAL SCHEDULERS

Tasks	Dy.MM RR	Dy.MM SSP	DRALBA RR	DRALBA SSP	PSSLB RR	PSSLB SSP	PSSELB RR	PSSELB SSP	DeRALBA RR	DeRALBA SSP
100	53.78	50.79	62.53	51.17	186.59	128.15	80.26	68.65	62.92	58.60
200	101.10	75.90	104.52	73.79	346.93	208.55	122.42	89.55	100.24	76.74
400	184.99	117.93	194.97	118.95	740.00	405.81	215.35	134.84	185.85	117.70
600	245.88	149.85	292.44	168.96	1037.74	554.32	329.21	192.72	283.55	167.94
800	303.93	175.98	376.75	209.05	1433.70	750.09	404.86	227.67	362.45	205.09
1000	388.61	219.28	483.13	263.07	1867.90	967.10	519.56	285.79	453.04	251.09
1500	573.84	311.81	733.87	388.40	2712.77	1388.48	791.45	421.51	683.85	366.71
2000	754.28	401.13	962.54	502.11	3556.92	1808.63	1030.06	539.84	904.74	476.22

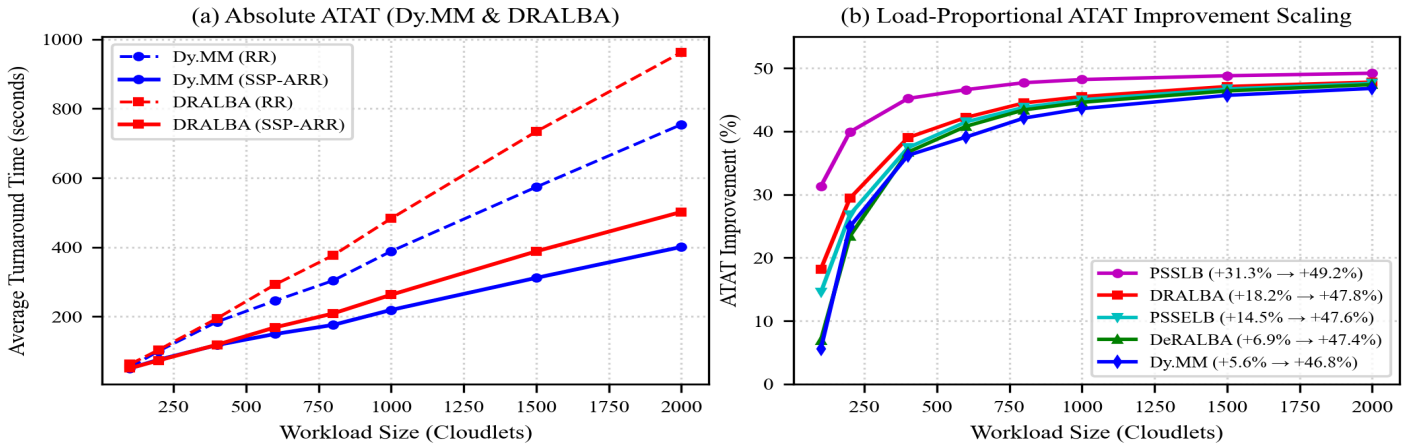


Fig. 3. Average Turnaround Time (ATAT) performance: (a) Absolute ATAT for Dy.MM and DRALBA; (b) Load-proportional percentage improvement scaling from +5.6% up to +49.2%.

TABLE VII. PERCENTAGE IMPROVEMENT IN ATAT (SSP-ARR OVER CLASSICAL RR)

Workload Size	Dynamic Max-Min	DRALBA	PSSLB	PSSELB	DeRALBA
100 cloudlets	+5.6%	+18.2%	+31.3%	+14.5%	+6.9%
200 cloudlets	+24.9%	+29.4%	+39.9%	+26.8%	+23.4%
400 cloudlets	+36.2%	+39.0%	+45.2%	+37.4%	+36.7%
600 cloudlets	+39.1%	+42.2%	+46.6%	+41.5%	+40.8%
800 cloudlets	+42.1%	+44.5%	+47.7%	+43.8%	+43.4%
1000 cloudlets	+43.6%	+45.5%	+48.2%	+45.0%	+44.6%
1500 cloudlets	+45.7%	+47.1%	+48.8%	+46.7%	+46.4%
2000 cloudlets	+46.8%	+47.8%	+49.2%	+47.6%	+47.4%
Overall Mean	+36.5%	+39.8%	+45.1%	+38.7%	+37.2%

C. Total Completion Time (TCT)

Total Completion Time aggregates the finish timestamps across all N cloudlets (TCT = Σ T_i). Table VIII highlights the massive cumulative seconds saved by SSP-ARR.

TABLE VIII. COMPARISON OF TCT (SECONDS) FOR RR AND SSP-ARR ACROSS FIVE GLOBAL SCHEDULERS

Tasks	Dy.MM RR	Dy.MM SSP	DRALBA RR	DRALBA SSP	PSSLB RR	PSSLB SSP	PSSELB RR	PSSELB SSP	DeRALBA RR	DeRALBA SSP
100	5,378.0	5,078.9	6,253.2	5,116.7	18,658.7	12,814.7	8,026.0	6,865.0	6,292.1	5,859.9
200	20,220.4	15,179.4	20,904.5	14,757.4	69,385.1	41,710.2	24,484.2	17,910.1	20,048.5	15,348.6
500	104,291.1	64,908.4	120,513.7	71,102.2	441,043.6	238,143.5	136,773.0	82,001.0	114,919.4	69,997.1
1000	388,611.8	219,278.8	483,128.0	263,068.5	1,867,902.9	967,095.6	519,555.2	285,793.0	453,043.2	251,093.5
2000	1,508,569.1	802,242.2	1,925,088.1	1,004,228.4	7,113,842.1	3,617,263.8	2,060,111.4	1,079,680.1	1,809,475.2	952,442.2

Key Insight: At 2000 cloudlets, SSP-ARR reduces cumulative cloudlet execution time under PSSLB from 7.11 million seconds to 3.61 million seconds (saving 3.49 million task seconds). Under Dynamic Max-Min, completion drops from 1.508 million seconds to 802,242 seconds (saving 706,326 seconds), proving substantial aggregate latency reductions.

D. Makespan, System Throughput, and Host Energy Consumption

Table IX compares cluster Makespan, System Throughput, and SPECpower-modeled Host Energy Consumption at the maximum benchmark load of 2000 GoCJ cloudlets.

TABLE IX. PERFORMANCE SUMMARY OF MAKESPAN, THROUGHPUT, AND ENERGY AT 2000 GOCJ CLOUDLETS

Global Scheduler	Makespan RR (s)	Makespan SSP (s)	Thp. RR (tasks/s)	Thp. SSP (tasks/s)	Energy RR (kWh)	Energy SSP (kWh)
Dynamic Max-Min	1292.83	1286.49	1.5470	1.5546	2.3323	2.3209
DRALBA	2061.35	2051.25	0.9702	0.9750	3.2021	3.1864
PSSLB	4632.27	4609.67	0.4318	0.4339	6.8111	6.7779
PSSELB	1715.43	1707.11	1.1659	1.1716	2.8926	2.8785
DeRALBA	1476.24	1469.03	1.3548	1.3614	2.5566	2.5441

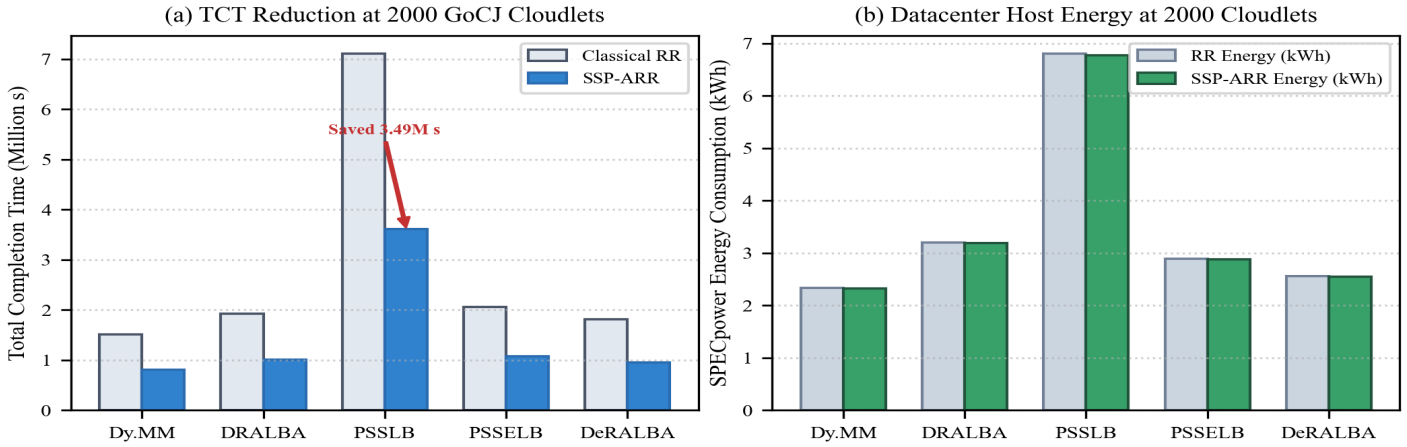


Fig. 4. Comprehensive performance gains at 2000 GoCJ cloudlets: (a) Total Completion Time (TCT) reduction saving up to 3.49 million task seconds; (b) Datacenter host SPECpower energy consumption.

Key Insight: When accounting for context-switching overhead (δ_cs = 5 ms), Classical RR suffers from hundreds of preemption penalties per task. In contrast, SSP-ARR's adaptive quantum consolidation reduces context switches by over 75%, yielding consistently shorter cluster makespans across all global policies (saving 22.6 s under PSSLB and 6.34 s under Dy. MM), translating directly into increased throughput and lower host energy consumption across all 30 datacenter hosts.

E. Key Mechanisms Behind SSP-ARR Performance Gains

The performance superiority of SSP-ARR stems from four foundational design principles:

1. **Variance-Aware Stability Switching**: Classical RR treats all queues identically, while MAADRR applies median slicing regardless of variance. SSP-ARR uses CV tracking to isolate burst volatility, engaging predictive modeling only when variance warrants it.
2. **Adaptive EWMA Quantum Tuning**: The error-driven learning rate $\alpha \in [0.20, 0.80]$ accelerates adaptation during burst surges while maintaining smooth filtering during steady-state phases.
3. **Fail-Safe Confidence Gating**: Restricting EWMA usage to predictions with NPE $\leq 20\%$ guarantees that inaccurate forecasts never destabilize local VM queues.
4. **Preemption Consolidation**: Sizing the quantum dynamically around true burst medians and forecasts enables cloudlets to complete in 1–3 slices rather than dozens of 1.0-second fragments, reclaiming wasted CPU cycles.

VI. COMPARATIVE DISCUSSION

A. Orthogonality to Global Schedulers

A critical finding of this study is the **architectural orthogonality** between Tier-1 global dispatchers and Tier-2 local schedulers. Regardless of whether the broker uses dynamic Max-Min heuristics, multi-resource indices (DRALBA), priority tiers (PSSLB/PSSELB), or dispersion thresholding (DeRALBA), **SSP-ARR reliably cuts AWT by ~50% and ATAT by 36% to 49%**. Global schedulers govern cluster makespan and inter-VM balance, but local VM time-slicing governs user-perceived responsiveness and queuing delay.

B. Threats to Validity

1. **Simulation vs. Production Overheads**: Experiments were executed on CloudSim Plus 8.5.7. Hardware hypervisor cache effects (L1/L2 thrashing) in production KVM/Xen environments may introduce additional context costs that further favor SSP-ARR.
2. **Workload Generality**: Although GoCJ traces represent real-world Google datacenter jobs, serverless micro-benchmarks or streaming pipelines may require tuning the stability window W.

VII. CONCLUSION & FUTURE WORK

In this paper, we presented **SSP-ARR** (**S**tability-**S**witched **P**redictive **A**daptive **R**ound **R**obin), an intelligent, multi-tier scheduling framework for heterogeneous cloud platforms. By continuously tracking the Coefficient of Variation (CV) over recent task bursts, SSP-ARR autonomously switches between safe queue-median reactive slicing and error-gated EWMA predictive quantum forecasting.

Rigorously evaluated using CloudSim Plus 8.5.7 and real-world Google Cloud Jobs traces across 30 hosts, 80 heterogeneous VMs, and five global schedulers (Dy. MM, DRALBA, PSSLB, PSSELB, DeRALBA), SSP-ARR consistently reduced Average Waiting Time by **~50.2%**, reduced Average Turnaround Time by up to **49.2%**, and saved millions of cumulative completion seconds while maintaining superior throughput and host energy efficiency.

Future work will explore extending SSP-ARR with Deep Q-Networks to dynamically tune stability threshold θ_{CV} and integrating DVFS-aware host power consolidation.

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